Trees 4 (DP on trees)

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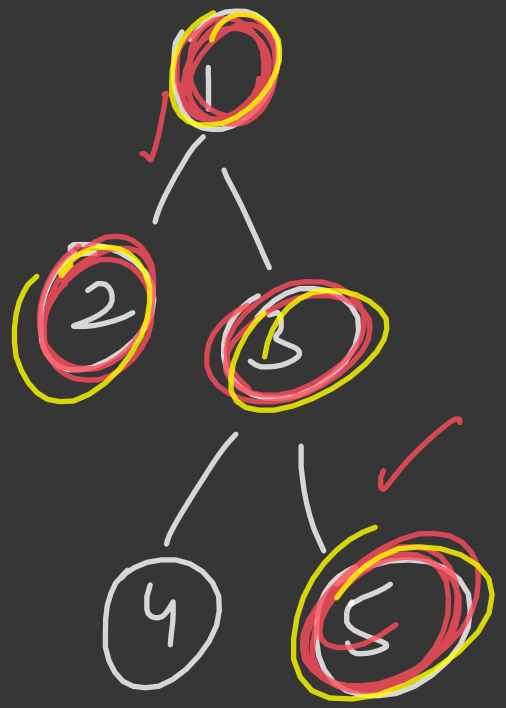
DP on Trees

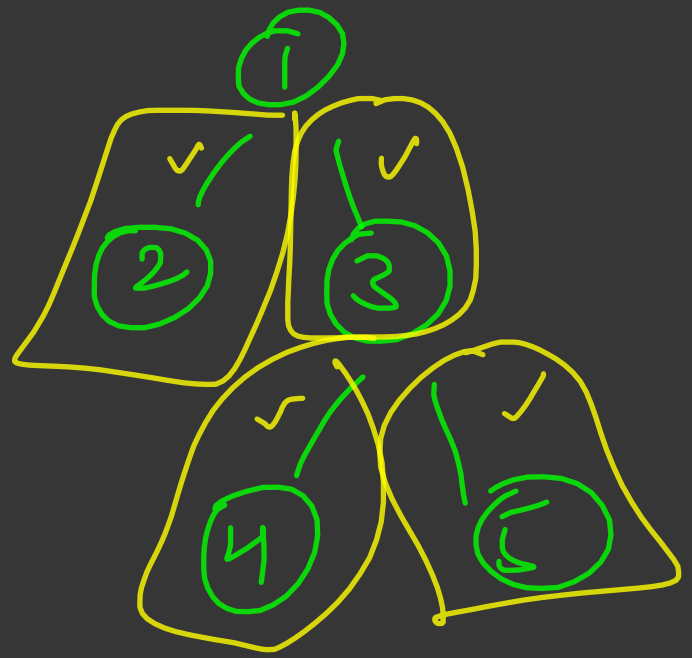
- General DP on trees
  - Tree Matching
- Re-rooting DP (UBER special)
  - Tree Distances 1
  - Tree Distances 2
  - Tree with Max Cost

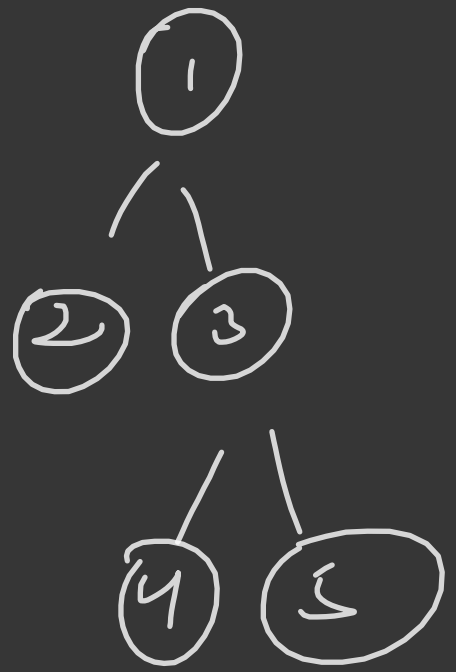
slice coding Test

1900 cf problem



2 answer





dp[1][0]

dp[node][flag]

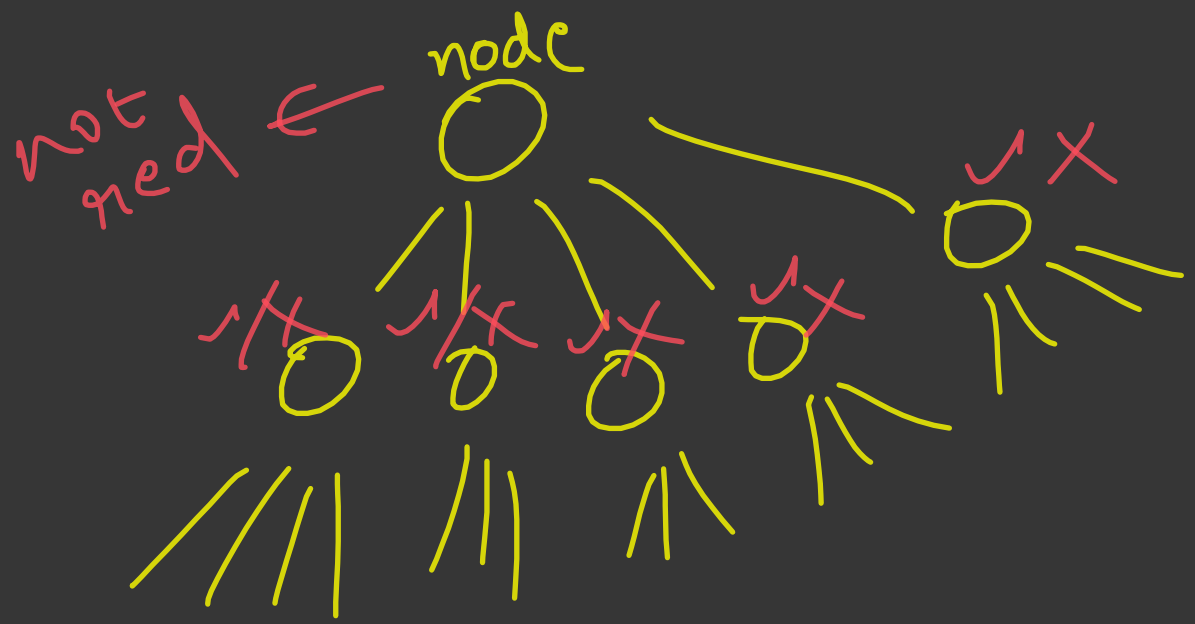
1 to n

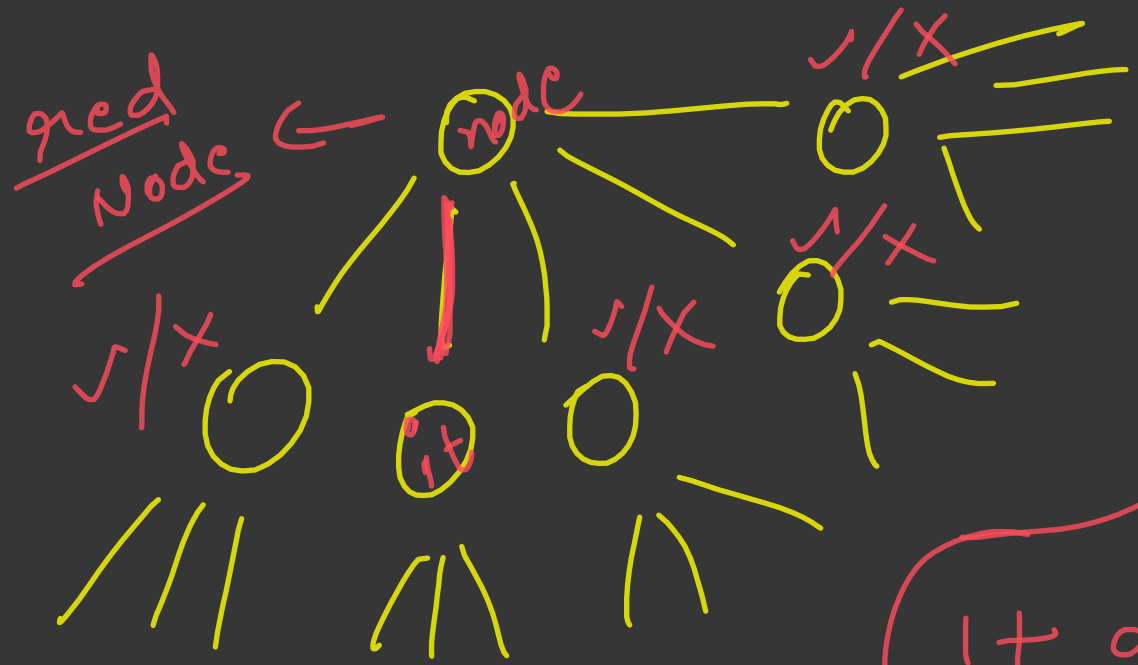
(0, 1)

dp[1][0] → 1 is not a red

dp[1][1] → 1 is a red node

$$dp[node][0] \mathrel{+}= \max(dp[it][0], dp[it][1])$$





$$dp[node][1] = \max\Big(dp[node][1],\ 1 + dp[node][0] - \max(dp[it][0], dp[it][1]) + dp[it][0]\Big)$$

$dp[1][0] = \max(dp[2][0], dp[2][1])$

$+$

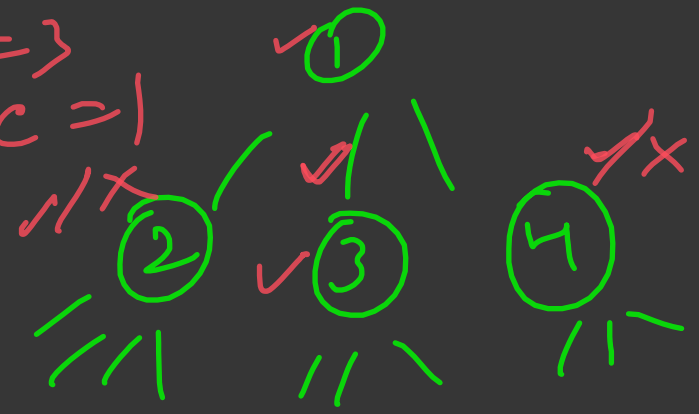
$\max(dp[3][0], dp[3][1])$

$+$

$\max(dp[4][0], dp[4][1])$

$it \Rightarrow$

$node = 1$



$dp[1][1] = \max(dp[1][1],$

$1 + dp[3][0]$

$\max(dp[2][0], dp[2][1]) +$

$\max(dp[4][0], dp[4][1])$

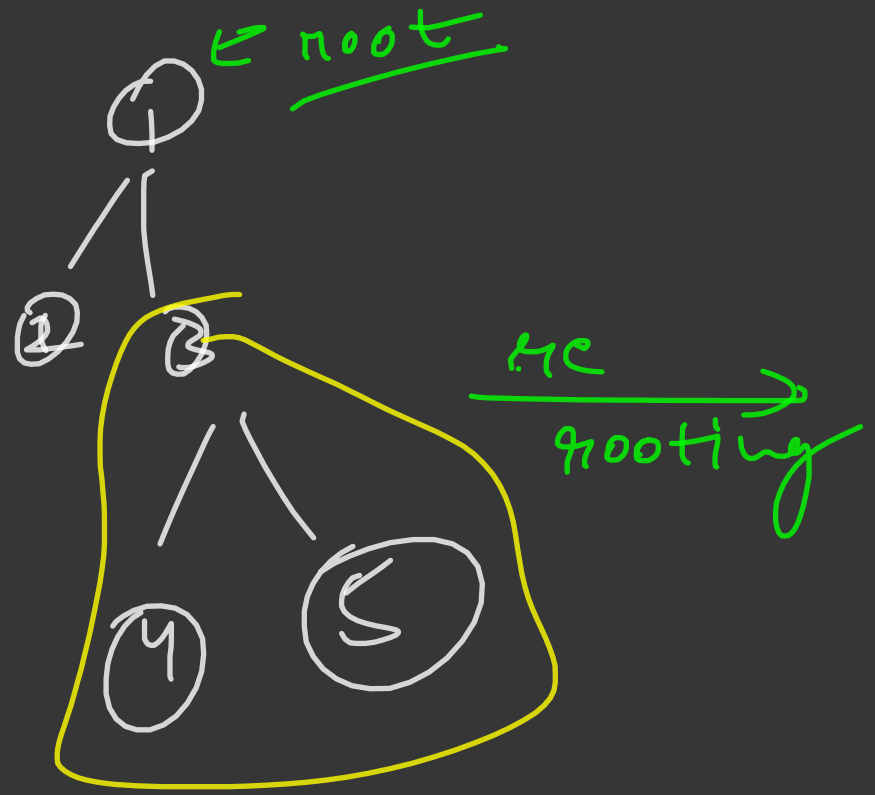
$TC \rightarrow O(V+E) \rightarrow O(n)$

$dp[1][0] - \max(dp[3][0], dp[3][1])$

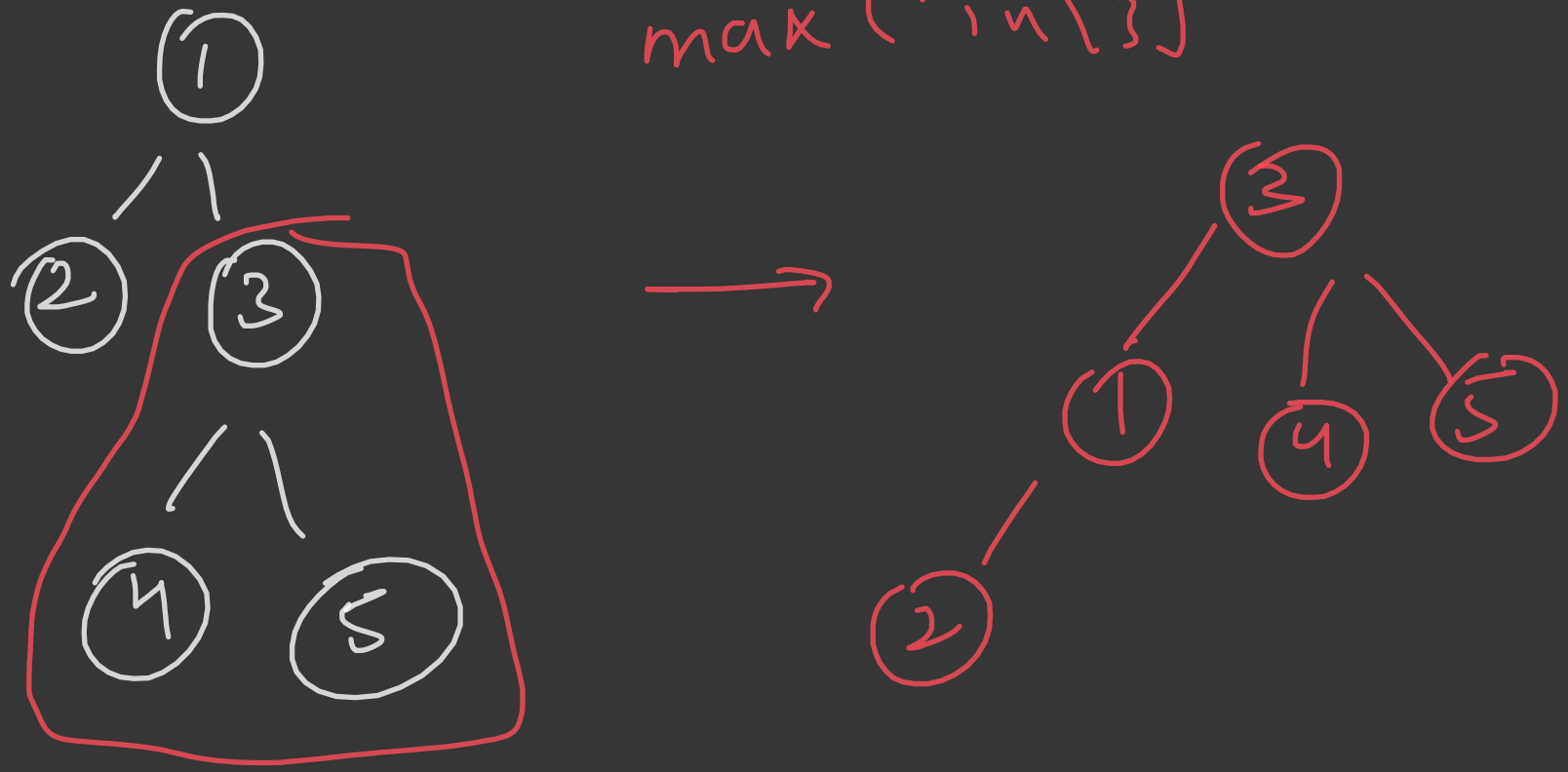
$$dp[\overset{node}{\ }][1] = \max(dp[\overset{node}{\ }][1], 1 + dp[it][0] + dp[\overset{node}{\ }][0] - \max(dp[it][0], dp[it][1]))$$

$$\max(dp[n][1], dp[n][0])$$

Re-rooting →



max(·in[3])



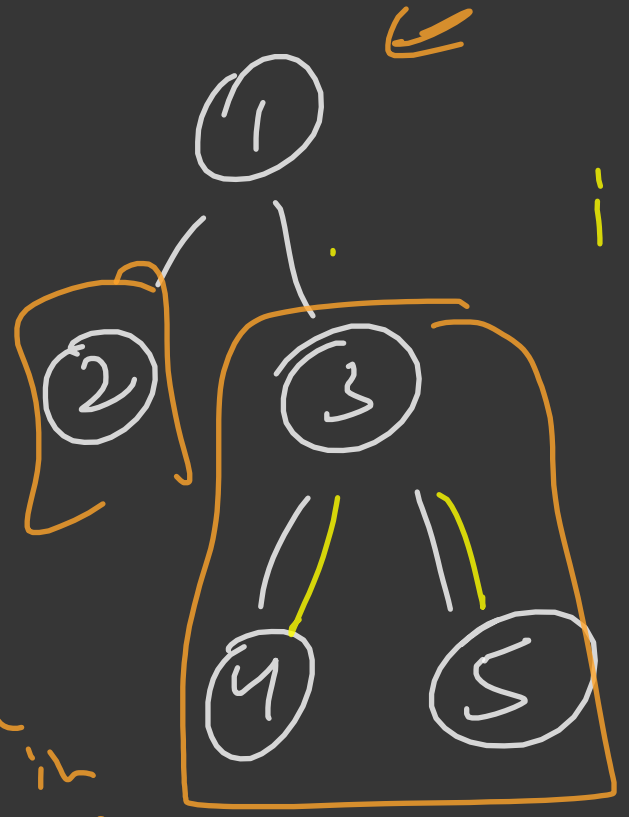
par
node

Q

in[node]

↓

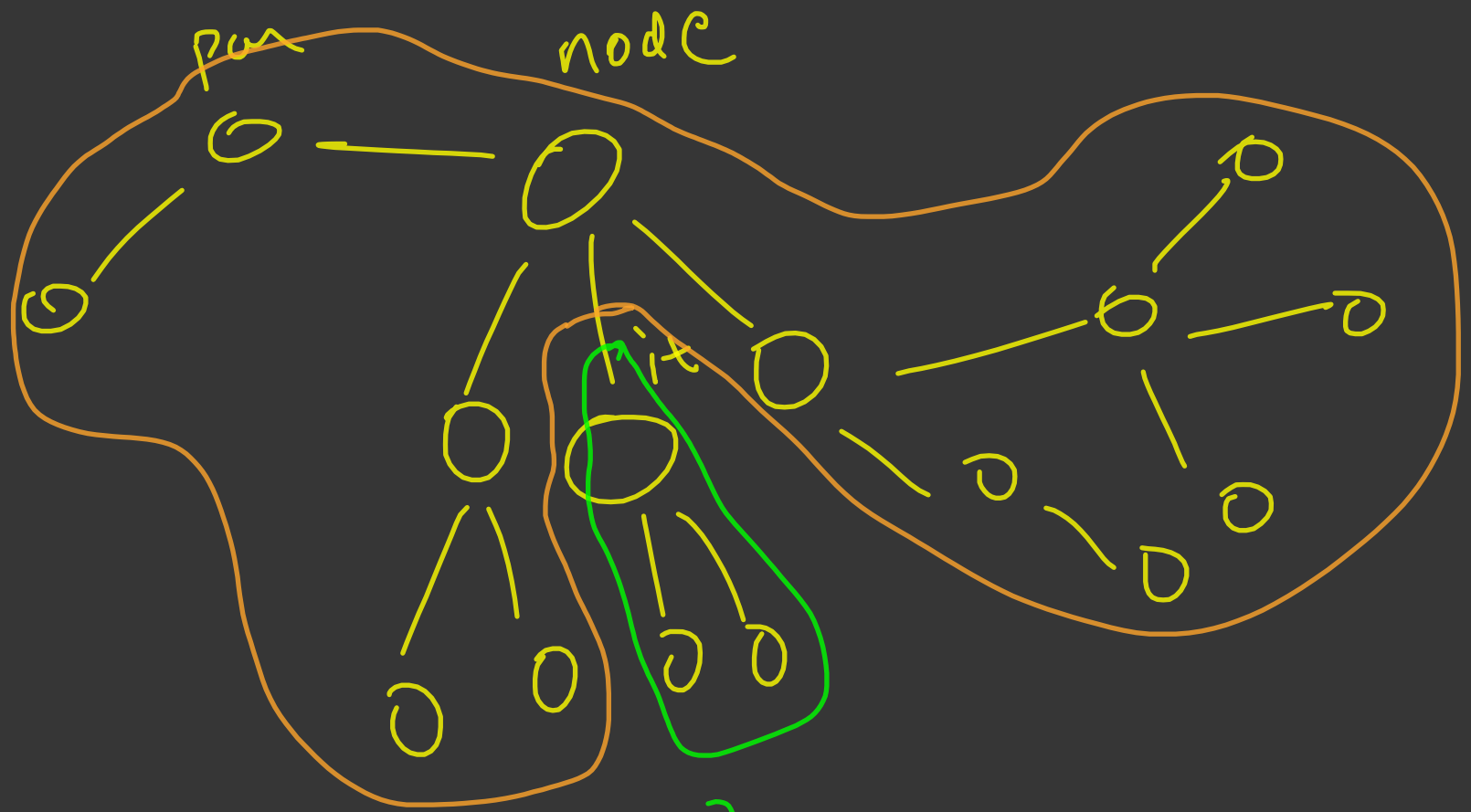
the sum of distance of each node in subtree



subtree[node]

in[node] += in[it] + subtree(it)

in[2]



$dp[it] = in[it] +$

$$dp[node] - \cancel{int[it]}$$

$$- subtree[it]$$

$$n - subtree[it]$$

$$\boxed{dp[it] = dp[node] + n - 2 \times subtree[it]}$$